

Progress Report Update 5: NSU ClubHub

A Centralized Club Management & Engagement Portal for North South University

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Abstract—This report presents the fifth milestone of NSU ClubHub, a centralized club management portal for North South University. This update completes the backend API development phase with Node.js, Express, Prisma ORM, and PostgreSQL. The system features JWT authentication, role-based access control, 25+ API endpoints, and complete frontend integration. All mock data has been replaced with dynamic API calls, completing the transition from frontend-only prototype to full-stack application.

Index Terms—Club Management, Backend API, Node.js, Express, Prisma, PostgreSQL, JWT Authentication, North South University

I. INTRODUCTION

NSU ClubHub is a web application enabling students to discover clubs, track recruitment, and stay updated on campus events. Update 5 transitions the system from frontend-only prototype to full-stack application with complete backend integration.

II. BACKEND ARCHITECTURE

Built using Node.js and Express.js with modular route-based design. Server runs on port 3001 with CORS enabled. JWT middleware protects routes with role-based authorization. Separate controllers handle authentication, clubs, events, recruitment, announcements, users, and dashboard statistics.

III. DATABASE SCHEMA (DB-01–DB-09)

Nine interconnected models using Prisma ORM: User (DB-01) stores accounts with email, password, name, student ID, role; Club (DB-02) maintains name, description, category, logo, cover image; ClubMember (DB-03) creates membership with role tracking; Event (DB-04) handles title, description, category, dates, venue, approval workflow; RSVP (DB-05) tracks event attendance; RecruitmentCycle (DB-06) manages recruitment with slot limits; Application (DB-07) stores position and status tracking; Announcement (DB-08) supports platform-wide notices; Notification (DB-09) provides user alerts.

IV. API ENDPOINTS (API-01–API-25)

25+ RESTful endpoints: Auth (register, login); Clubs (CRUD with filters); Events (CRUD with RSVP); Recruitment (cycles and applications); Dashboard (role-specific statistics).

V. AUTHENTICATION SYSTEM

JWT-based authentication with bcrypt. Three roles: STUDENT, CLUB_ADMIN, SUPER_ADMIN. Protected routes verify tokens and check roles. Test accounts: admin@nsu.edu (Super Admin), arif@nsu.edu (Club Admin), student@nsu.edu (Student).

VI. FRONTEND INTEGRATION

Next.js frontend connected to API using React Context and hooks. AuthContext manages global state. api.ts provides typed endpoint functions. All dashboards fetch live data.

VII. CONCLUSION

Update 5 completes the backend phase with 25+ endpoints, secure authentication, and database schema. Frontend integration enables dynamic data flow. Future work includes image upload, email notifications, and production deployment.

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